ACIT-HUB

ACCELERATED COMPUTE INFRASTRUCTURE TRAINING HUB

PERFORMANCE MONITORING HACKATHON

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UK Research and Innovation
Engineering and Physical Sciences Research Council

The ACIT Hub

- A **UKRI DRI Training Hub** for Research Technology Professionals, backed by **£3.9m** of funding
- Builds the profession of the **Research Infrastructure Engineer (RIE)** — the specialists who keep digital research infrastructure running
- Delivers **industry-trusted training**, promotes best practice and raises the visibility of RIE careers
- Responds to **rising AI demands** on research facilities and the teams behind them

Hub partner institutions

University of Surrey
University of Bristol
Imperial
University of Oxford
University of Sheffield

Industry partners

Hewlett Packard Enterprise
Dell Technologies
AMD
NVIDIA
WEKA

bcs The Chartered Institute for IT
Software Sustainability Institute
BMVA
HPC SIG The High Performance Computing Special Interest Group for UK Academia
NPL

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Hackathon

A one-day sprint exploring GPU efficiency and performance monitoring

25

engineers

16

universities & institutes

1

1 day, hands-on, 21st April

WHO

Research Infrastructure Engineers, Research Software Engineers, and users of accelerated compute

HANDS-ON WITH

Grafana · CEEMS · Prometheus, building performance monitoring dashboards from scratch

Why Performance Monitoring?

Cost & Utilisation

GPUs are the cluster's most expensive resource, and also its most under-utilised. Even a 10% lift in utilisation pays back the engineering effort many times over.

Energy & Sustainability

GPU power draw dominates HPC energy bills. Per-job energy accounting via CEEMS exposes idle-but-powered nodes and supports funder reporting on kWh and CO_2.

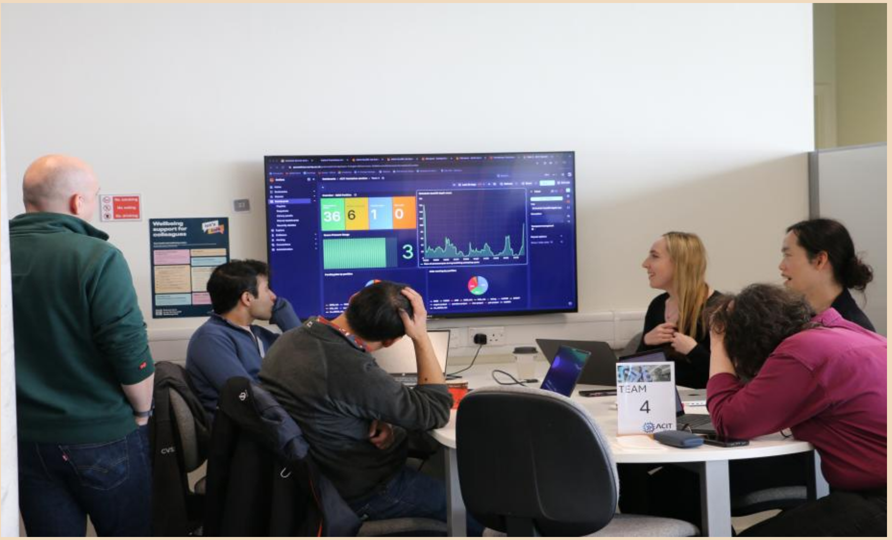
Evidence-Based Operations

Turn "the cluster feels slow" tickets into 30-second diagnoses (thermal throttling, NVLink errors, noisy neighbours), and justify the next procurement with hard data.

User-Facing Transparency

Exposing job-level efficiency back to researchers changes behaviour without RIEs having to police it. Users tune their own workloads once they can see them.

From idea to dashboard in eight hours



1. **WELCOME**
   Kick-off and intros
2. **TEAM UP**
   Groups of 4–5 RIEs & RSEs
3. **IDEATE**
   Brainstorm useful dashboards
4. **BUILD**
   Prototype on live cluster data
5. **SHARE**
   Show & tell of learnings

Tools we explored

CEEMS

EXPORT

Collects per-job energy, power and efficiency metrics from each node.

Prometheus

COLLECT & STORE

Scrapes metrics from CEEMS and other exporters into a time-series database.

Grafana

VISUALISE

Turns the stored metrics into dashboards RIEs and users can read at a glance.

Open source · Widely used across UK and EU research computing

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What teams built

TEAM 2 · CLUSTER UTILISATION OVERVIEW

Aggregated cluster overview

An at-a-glance view of overall cluster utilisation, intended to give admins and PIs a single dashboard for everyday health and load.

TEAM 3 · NODE & PARTITION USAGE

Node, GPU and partition usage

Aimed at management: how heavily individual nodes, GPU types and partitions are used over a given time window, to inform which nodes to retire and which to focus on for new procurement.

What worked & why?

Tailoring dashboards to the audience	Teams designed for distinct users (beginner cluster users, advanced users, PIs, and cluster admins) so each dashboard stayed focused on a specific context and this produced diverse dashboard designs.
Starting from real cluster data	Working with live metrics from several sites, not synthetic test data, meant teams got immediate signal on what was useful, and prototypes were directly portable back to their home clusters.
Small teams of 4–5 people	Big enough to bring complementary RIE and RSE skills to each prototype, small enough that everyone contributed and no one was a passenger.
A one-day, time-boxed format	Tight enough to keep energy and focus high without losing a working week. It encouraged teams to ship something rough rather than over-plan.
Breakout booths and a central area	The venue gave teams space to concentrate in small groups, then bring everyone back together for talks and cross-pollination throughout the day.

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Challenges

Access and permissions

CHALLENGE

Setting up participant access to metrics and Wi-Fi credentials ate into the morning. Some job data needed elevated permissions that aren't trivial to grant.

LESSON LEARNED

Stage credentials and access tiers ahead of time; brief participants the day before so day one starts with building, not paperwork.

Brainstorming bottleneck

CHALLENGE

Example dashboards were provided, but several groups struggled to converge on a single idea and lost momentum early in the day.

LESSON LEARNED

Offer a short menu of starter problems teams can adopt as-is, with freedom to depart from them. This reduces decision fatigue without limiting creativity.

Getting to grips with CEEMS data

CHALLENGE

Building useful visualisations required familiarity with the CEEMS metric structures. Teams needed time to feel out the data before they could start charting.

LESSON LEARNED

Run a short, focused walkthrough of the CEEMS data model up front so teams hit the ground charting, not exploring.

Embedded experts spread too thin

CHALLENGE

Grafana and Prometheus experts were assigned to specific groups but kept being pulled across to help others, leaving their host group under-served.

LESSON LEARNED

Run experts as floating mentors with a posted rotation rather than group members. Same expertise, better coverage, and no group owns the expert.

Feedback

WHAT I LEARNED

"Monitoring is important to different stakeholders, and each requires a different set of data and dashboards."

"The variety of metrics that can be combined to produce informative output in Grafana dashboards."

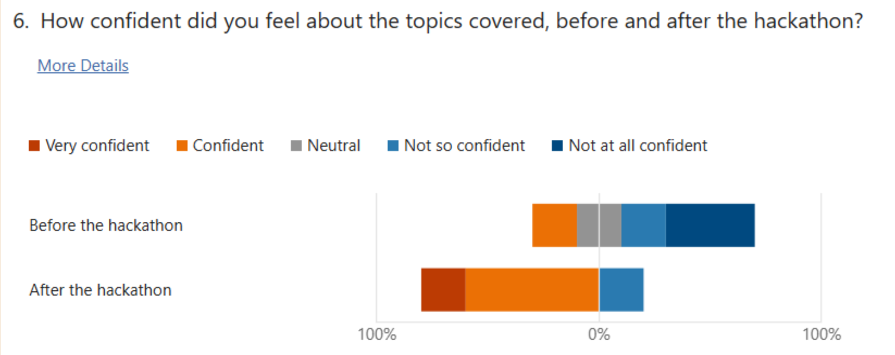
WHAT I LIKED

"Dedicated time off from work to learn something new, networking, learning from other people, and working as a team."

WHAT I'LL DO NEXT

"Improve performance monitoring of our GPUs and explore using CEEMS."

"Create a dashboard."



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Join us next time

14 JUL	AUG / SEP	15 SEP	16 SEP	7 OCT
WEBINAR	WEBINAR	HACKATHON	CO-DESIGN DAY	WORKSHOP
Agentic Ops	**Hackathon Prep**	**Same event, two places**	**University of Sheffield**	**Agile for RIE's**
Richard Gilham, Isambard		Surrey + Sheffield	3 more in November. Edinburgh, Oxford & Bristol	University of Surrey

Join us, get involved

Register and find more at the ACIT Hub website, or join the mailing list to hear about future events.

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UK Research and Innovation

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